

Course Title	Analogue & Digital Communication	Course Code	EE3003
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Research Literature (WK8)	Credit Hrs.	3+1
Knowledge Area	Telecommunication (KA12)	Grading Scheme	Relative
HEC Knowledge Area	Electrical Engineering Core (Breadth)	Applicable From	Fall 2023
Pre-requisite(s)	EE/EL2008 Signals and Systems, EE2011 Probability and Random Processes		

Course Objective	To get acquainted with the concept of communication and become familiar with mathematical handling of power and energy signals, time and frequency axis, multiplexing and modulation concepts along with some system performance parameters.
-------------------------	--

No.	Assigned Program Learning Outcome (PLO)
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Recognize the parameters of a communication system and the noise/distortion therein.	A1,M1	C1	2
2	Compute energy and power spectral density of the signal	Q1,M1	C3	2
3	Demonstrate the concept of analog amplitude modulation and demodulation techniques	A2,M1,M2	C3	2
4	Demonstrate the concept of analog frequency modulation and demodulation techniques	Q2,M2	C3	2
5	Demonstrate analog signal to digital signal conversion with encoding and Interpret demodulation/detection of digital signals	A3,Q3,F	C3	2

Text Book(s)	Title	Modern Digital and Analog Communication Systems, 3rd Edition
	Author	B P Lathi
	Publisher	Oxford University Press
Ref. Book(s)	Title	Fundamentals of Communication Systems, 2nd Edition, 2014
	Author	John G. Proakis, Masoud Salehi
	Publisher	Pearson Education Inc. (Prentice Hall)
	Title	Digital Communications, Fundamentals and Applications, 2nd Ed.
	Author	Bernard Sklar
	Publisher	Prentice Hall PTR
	Title	Communication Systems, 4rd Edition
	Author	Simon Haykin
	Publisher	John Wiley and Sons

Week	Course Contents/Topics	Chapter*	CLO*
1	Introduction to communication systems and representation of aperiodic signals	1, 3	1
2	Signal transmission through a linear system, Signal distortion over a communication channel	3	1
3	Measure of signals, ESD and PSD of modulated signals	3	2
4	Analogue Techniques: Double side band (DSB) amplitude modulation, AM	4	3
5	Bandwidth-efficient AM (SSB, QAM, VSB)	4	3
6	Local carrier synchronization in suppressed carrier schemes, frequency division multiplexing (FDM), PLL and some applications	4	3
7	Concept of instantaneous frequency, bandwidth of angle-modulated waves	5	4
8	Generation of FM waves (indirect and direct methods), demodulation of FM waves	5	4
9	Effects of non-linear distortion and interference, Superheterodyne analog AM/FM Receivers and FM broadcasting system	5	4
10	Digital Techniques: Sampling theorem (signal reconstruction using samples with practical issues and some applications of sampling theorem)	6	5
11	Baseband Techniques: PAM, PWM, PPM, Pulse Code Modulation (PCM)	6	5
12	Differential PCM (DPCM), Delta modulation (DM), Adaptive DM	6	5
13	Digital communication system, Line coding (polar signaling, on-off signaling, Bipolar Signaling)	1,7	5
14	Pulse Shaping, Nyquist criteria for zero ISI	7	5
15	Scrambling, Regenerative repeater	7	5

*Reference book chapters are given in brackets

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterms (I+II)	15% +15%
Final Exam	50.0%